

Shortest path algorithm

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Practice quiz

The shortest path algorithm, to calculate the shortest paths from node o to all the other nodes in the network, is described as follows.

1. Initialization: $\mathcal{S} \leftarrow \{o\}$, $\lambda_o = 0$, $\lambda_j = +\infty$.
2. At each iteration:
 - Select i in $\mathcal{S}$ and remove it from $\mathcal{S}$.
 - For each (i, j) :
 - If $\lambda_j > \lambda_i + c_{ij}$,
 - * $\lambda_j \leftarrow \lambda_i + c_{ij}$.
 - * If $\lambda_j < 0$ and $\lambda_j < (m - 1) \min_{(i,j) \in \mathcal{A}} c_{ij}$: STOP.
 - * $\mathcal{S} \leftarrow \mathcal{S} \cup \{j\}$.
 - If $\mathcal{S} = \emptyset$: STOP.
 - Otherwise, start again.

Apply the algorithm to identify the shortest paths from node 1 to all other nodes in the following network, and identify the shortest path tree.

